

Credit Card Fraud Detection Report

The Credit Card Fraud Analysis dashboard provides an overview of transaction activity and fraud trends across different card types, authentication methods, transaction channels, device types, and time periods. The objective of the analysis is to identify fraud patterns that can help financial institutions strengthen fraud detection and reduce financial losses.

Key Performance Indicators (KPIs)

Metric	Value
Total Transactions	20,000
Fraud Rate	2%
Total Fraud Value	61.44K
Average Legitimate Transaction Amount	131.58
Average Fraudulent Transaction Amount	181.25

Interpretation

- The dataset contains **20,000 transactions**.
- The overall fraud rate is approximately **2%**, indicating that fraudulent transactions represent a small but significant portion of all transactions.
- Fraudulent transactions resulted in a total loss of **61.44K**.
- The average fraudulent transaction amount (**181.25**) is higher than the average legitimate transaction amount (**131.58**), suggesting that fraudsters tend to target higher-value transactions.

➤ Fraud Rate by Card Type

The dashboard compares fraud rates across different card providers.

Findings

- **Amex** records the highest fraud rate.
- **Discover** follows closely behind.
- **Visa** experiences a moderate fraud rate.

- **Mastercard** has a lower fraud rate than Visa.
- **Ru Pay** records the lowest fraud rate among the card types analysed.

Business Insight

Premium cards such as **Amex** appear to be more attractive targets for fraudsters due to their generally higher spending limits and transaction values.

➤ **Fraud Rate by Authentication Method**

The dashboard evaluates fraud rates based on authentication methods.

Findings

- Transactions with **No Authentication** have the highest fraud rate.
- **OTP** and **3D Secure** significantly reduce fraud.
- **PIN** authentication performs better than having no authentication.
- **Biometric authentication** records the lowest fraud rate.

Business Insight

Strong authentication methods substantially reduce fraud risk. Encouraging customers to use biometric authentication or multi-factor authentication can improve transaction security.

➤ **Fraud Rate by Channel and Device Type**

Fraud rates remain relatively consistent across channels and devices.

Findings

- Fraud rates are similar across:
 - ATM
 - Contactless
 - In-App
 - Online
 - POS

- Device types including Android phones, iPhones, Macs, ATMs, and POS terminals all record fraud rates close to **2%**.

Business Insight

Fraud is not concentrated within a single transaction channel or device, indicating that fraud prevention measures should be applied consistently across all platforms.

➤ **Foreign vs Domestic Fraud**

The dashboard compares foreign and domestic fraudulent transactions.

Findings

- Foreign fraud accounts for approximately 80% of fraud cases.
- Domestic fraud contributes around 20%.

Business Insight

International transactions present significantly higher fraud risk. Additional verification should be applied to overseas transactions.

➤ **AI Scam Fraud Analysis**

The dashboard distinguishes between AI-related scams and non-AI scams.

Findings

- Around **85%** of fraud cases are AI scam-related.
- Approximately **15%** are non-AI scams.

Business Insight

AI-assisted fraud is becoming increasingly prevalent. Financial institutions should invest in AI-driven fraud detection systems capable of identifying sophisticated scam patterns.

➤ **Fraud Cases by Time of Day**

Fraud occurrence varies throughout the day.

Findings

- **Early Morning** experiences the highest number of fraud cases.
- Morning, Afternoon, and Evening record noticeably fewer fraud incidents.

Business Insight

Fraudsters may exploit periods of reduced customer monitoring and lower operational oversight during early morning hours. Banks should increase real-time monitoring during these periods.

Overall Insights

The analysis reveals several important trends:

- Fraudulent transactions generally involve higher monetary values than legitimate transactions.
- Amex cards experience the highest fraud rate.
- Transactions without authentication are considerably more vulnerable to fraud.
- Foreign transactions contribute the majority of fraudulent activity.
- AI-enabled scams account for most fraud cases.
- Early morning is the period with the highest fraud occurrence.
- Fraud rates remain relatively consistent across transaction channels and device types.

Recommendations

Based on the analysis, the following actions are recommended:

1. Implement mandatory multi-factor authentication for high-risk transactions.
2. Increase monitoring of international transactions using real-time fraud detection systems.
3. Deploy AI-based fraud detection models capable of identifying emerging scam patterns.

4. Apply additional verification checks for high-value transactions.
5. Strengthen fraud monitoring during early morning hours when fraud activity peaks.
6. Continuously educate customers about AI-enabled scams, phishing, and social engineering attacks.
7. Regularly review fraud patterns by card type to improve risk-based security controls.

Conclusion

The dashboard indicates that although the overall fraud rate is relatively low at **2%**, fraudulent transactions generate substantial financial losses and are concentrated in specific risk areas. The highest risks are associated with **Amex cards, unauthenticated transactions, foreign transactions, AI-enabled scams, and early morning activity**. By strengthening authentication methods, enhancing AI-driven monitoring, and focusing on high-risk periods and transaction types, organisations can significantly reduce fraud exposure and improve customer security.